1. A monochromatic light wave with wavelength λ_1 and frequency ν_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency ν_2 of the refracted wave are:

[2023 (06 Apr Shift 1)]

(1) $\lambda_2=\sqrt{2}\lambda_1, \nu_2=\nu_1$

(2) $\lambda_2=\lambda_1, \nu_2=\frac{1}{\sqrt{2}}\nu_1$

(3) $\lambda_2=\lambda_1, \nu_2=\sqrt{2}\nu_1$

(4) $\lambda_2=\frac{1}{\sqrt{2}}\lambda_1, \nu_2=\nu_1$

2. A pole is vertically submerged in swimming pool, such that it gives a length of shadow $2.15 \mathrm{~m}$ within water when sunlight is incident at an angle of 30° with the surface of water. If swimming pool is filled to a height of $1.5 \mathrm{~m}$, then the height of the pole above the water surface in centimeters is $\left(n_w=\frac{4}{3}\right)$ ________.

[2023 (06 Apr Shift 1)]

3. A 2 meter long scale with least count of $0.2 \mathrm{~cm}$ is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at $80 \mathrm{~cm}$ mark and $1 \mathrm{~m}$ mark, respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at $180 \mathrm{~cm}$ mark. The $\%$ error in the estimation of focal length is:

[2023 (06 Apr Shift 2)]

(1) 0.85

(2) 1.70

(3) 1.02

(4) 0.51

4. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: The phase difference of two light waves change if they travel through different media having same thickness, but different indices of refraction.

Reason R: The wavelengths of waves are different in different media.

In the light of the above statements, choose the most appropriate answer from the options given below

[2023 (06 Apr Shift 2)]

(1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**

(2) **A** is not correct but **R** is correct

(3) **A** is correct but **R** is not correct

(4) Both **A** and **R** are correct and **R** is the correct explanation of **A**

5. In a reflecting telescope, a secondary mirror is used to:

[2023 (08 Apr Shift 1)]

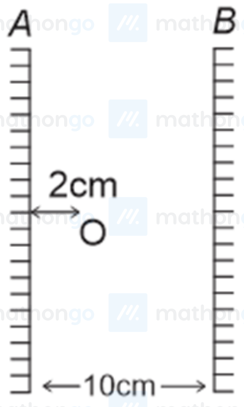
(1) reduce the problem of mechanical support

(2) make chromatic aberration zero

(3) move the eyepiece outside the telescopic tube

(4) remove spherical aberration

6. Two vertical parallel mirrors A and B are separated by 10 cm. A point object O is placed at a distance of $2 \mathrm{~cm}$ from mirror A. The distance of the second nearest image behind mirror A from the mirror A is _______ cm.



[2023 (08 Apr Shift 1)]

7. Two transparent media having refractive indices 1.0 and 1.5 are separated by a spherical refracting surface of radius of curvature $30 \mathrm{~cm}$. The centre of curvature of surface is towards denser medium and a point object is placed on the principal axis in rarer medium at a distance of $15 \mathrm{~cm}$ from the pole of the surface. The distance of image from the pole of the surface is cm.

[2023 (08 Apr Shift 2)]

8. An object is placed at a distance of 12 cm in front of a plane mirror. The virtual and erect image is formed by the mirror. Now the mirror is moved by 4 cm towards the stationary object. The distance by which the position of image would be shifted, will be

[2023 (10 Apr Shift 1)]

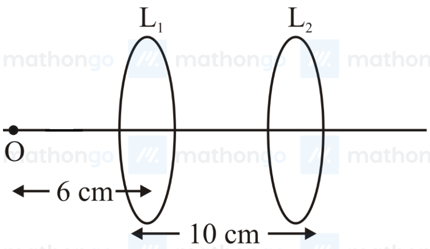
(1) 4 cm towards mirror

(2) 8 cm towards mirror

(3) 8 cm away from mirror

(4) 2 cm towards mirror

9. A point object O is placed in front of two thin symmetrical coaxial convex lenses L_1 and L_2 with focal length 24 cm and 9 cm respectively. The distance between two lenses is 10 cm and the object is placed 6 cm away from lens L_1 as shown in the figure. The distance between the object and the image formed by the system of two lenses is _____________ cm



[2023 (10 Apr Shift 2)]

10. The critical angle for a denser-rarer interface is 45°. The speed of light in rarer medium is 3×10^8 m s^{-1}. The speed of light in the denser medium is:

[2023 (11 Apr Shift 1)]

(1) 3.12×10^7 m s^{-1}

(2) 5×10^7 m s^{-1}

(3) 2.12×10^8 m s^{-1}

(4) $\sqrt{2} \times 10^8$ m s^{-1}

11. The radius of curvature of each surface of a convex lens having refractive index 1.8 is 20 cm. The lens is now immersed in a liquid of refractive index 1.5. The ratio of power of lens in air to its power in the liquid will be $x : 1$. The value of x is

[2023 (11 Apr Shift 1)]

12. When one light ray is reflected from a plane mirror with 30° angle of reflection, the angle of deviation of the ray after reflection is:

[2023 (11 Apr Shift 2)]

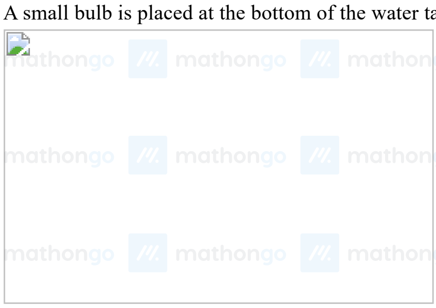
(1) 120°

(2) 110°

(3) 140°

(4) 130°

13. As shown in the figure, a plane mirror is fixed at a height of 50 cm from the bottom of tank containing water $\left(\mu = \frac{4}{3}\right)$. The height of water in the tank is 8 cm. A small bulb is placed at the bottom of the water tank. The distance of image of the bulb formed by mirror from the bottom of the tank is ______ cm.



[2023 (11 Apr Shift 2)]

14. An ice cube has a bubble inside. When viewed from one side the apparent distance of the bubble is 12 cm. When viewed from the opposite side, the apparent distance of the bubble is observed as 4 cm. If the side of the ice cube is 24 cm, the refractive index of the ice cube is

[2023 (12 Apr Shift 1)]

(1) $\frac{3}{2}$

(2) $\frac{2}{3}$

(3) $\frac{6}{5}$

(4) $\frac{4}{3}$

15. Two convex lenses of focal length $20\ \text{cm}$ each are placed coaxially with a separation of $60\ \text{cm}$ between them. The image of the distant object formed by the combination is at _____ cm from the first lens.

[2023 (12 Apr Shift 1)]

16. A vessel of depth d is half filled with oil of refractive index n_1 and the other half is filled with water of refractive index n_2. The apparent depth of this vessel when viewed from above will be-

[2023 (13 Apr Shift 1)]

(1) $\frac{2d(n_1+n_2)}{n_1+n_2}$

(2) $\frac{d(n_1+n_2)}{2n_1n_2}$

(3) $\frac{dn_1n_2}{2(n_1+n_2)}$

(4) $\frac{dn_1n_2}{(n_1+n_2)}$

17. A fish rising vertically upward with a uniform velocity of $8\ \text{m s}^{-1}$, observes that a bird is diving vertically downward towards the fish with the velocity of $12\ \text{m s}^{-1}$. If the refractive index of water is $\frac{4}{3}$, then the actual velocity of the diving bird to pick the fish, will be__________m s^{-1}.

[2023 (13 Apr Shift 1)]

18. A bi convex lens of focal length $10\ \text{cm}$ is cut in two identical parts along a plane perpendicular to the principal axis. The power of each lens after cut is _____ D .

[2023 (13 Apr Shift 2)]

19. The refractive index of a transparent liquid filled in an equilateral hollow prism is $\sqrt{2}$. The angle of minimum deviation for the liquid will be _______$^{\circ}$.

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (4)	**2.** (50)	**3.** (2)	**4.** (4)	**5.** (3)	**6.** (18)	**7.** (30)	**8.** (2)
9. (34)	**10.** (3)	**11.** (4)	**12.** (1)	**13.** (98)	**14.** (1)	**15.** (100)	**16.** (2)
17. (3)	**18.** (5)	**19.** (30)					

1. (4)

By using Snell's law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, the refractive index is

$\mu = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$

Also using the relation $n = \frac{c}{v}$ where n is the refractive index, c is the velocity of light in air and v is the velocity in the medium

$\Rightarrow \frac{c_{\text{air}}}{\text{v}} = \sqrt{2} = \frac{\lambda_1}{\lambda_2}$

The velocities are given by

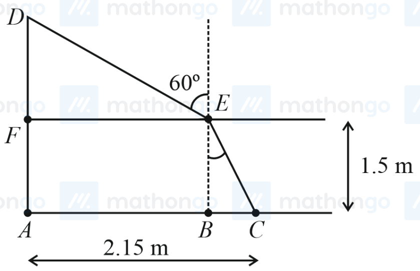
$v = \nu \lambda_2$

$c = \nu \lambda_1$

$\Rightarrow \lambda_1 = \sqrt{2} \lambda_2$ and $\nu_1 = \nu_2$.

The frequency of light does not change when the medium changes.

2. (50)



Let angle BAC is θ.

Applying Snell's law at E, we get $1 \times \sin 60^\circ = \frac{4}{3} \times \sin \theta$

$\Rightarrow \sin \theta = \frac{3\sqrt{3}}{8}$

$\Rightarrow \tan \theta = \frac{\sin \theta}{\sqrt{1 - (\sin \theta)^2}} = \frac{3\sqrt{3}}{\sqrt{37}}$

Now, $BC = 1.5 \times \tan \theta = 1.28$

Then, $AB = 2.15 - BC = 0.86 \text{ m}$

Required value, $DF = AB \tan 30^\circ = 0.5015 \text{ m} \approx 50 \text{ cm}$

3. (2)

The formula for a lens is given by $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \ldots (i)$

$u = (100 \pm 0.2) \text{ cm} - (80 \pm 0.2) \text{ cm} = (20 \pm 0.4) \text{ cm}$

$v = (180 \pm 0.2) \text{ cm} - (100 \pm 0.2) \text{ cm} = (80 \pm 0.4) \text{ cm}$

Substituting the values in the equation of focal length,

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{80} - \left(-\frac{1}{20}\right) = \frac{5}{80}$

$\Rightarrow f = 16 \text{ cm}$

Differentiating the equation (i)

$\frac{df}{f^2} = \frac{dv}{v^2} + \frac{du}{u^2}$(Only positive sign is considered for error calculation)

$= \frac{0.4}{6400} + \frac{0.4}{400} = 0.4 \times \frac{17}{6400}$

$\Rightarrow df = 0.4 \times \frac{17}{6400} \times 16^2 = 0.272$

The percentage error is

$\frac{df}{f} \times 100 = \frac{0.272}{16} \times 100 = 1.70\%$

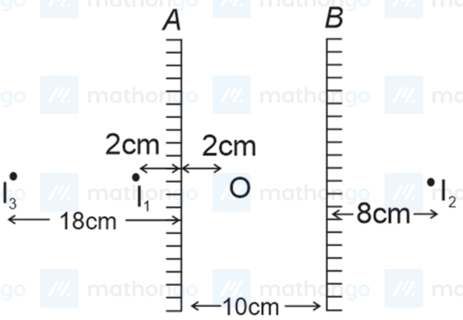
4. (4)

When light travels through a medium, its velocity depends on the density of the medium.
The formula to calculate the speed of light can be written as
$v=\nu\lambda \quad \ldots(1)$
From equation (1), it can be concluded that higher the speed, higher is the wavelength. Hence, the wavelength of light depends on the refractive index of the medium.
Because of changed wavelength the phase difference changes while the two waves travel the same distance.
Thus, both **A** and **R** are correct and **R** is the correct explanation of **A**.

5. (3)

Mirrors are used in reflecting telescopes, which allow astronomers to see distant objects in space more clearly.
A mirror forms an image by gathering light from spatial objects. A second mirror receives the picture that was reflected by the first mirror, which may be rather wide. The light is reflected by this tiny mirror onto an eyepiece lens, which magnifies or enlarges the image of the object.
Hence, the secondary mirror is used to move the eyepiece outside the telescopic tube.

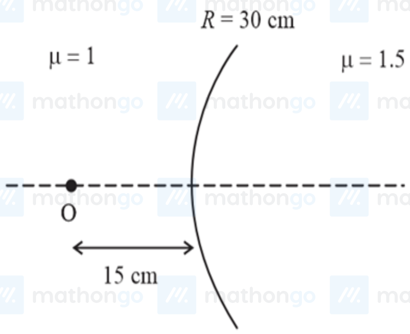
6. (18)



As can be seen from the image above, object distance is $8\ \text{cm}$ [For image by B]
In a plane mirror the image distance is equal to the object distance. Thus, the required distance is
$8+8+2\ \text{cm}=18\ \text{cm}$

7. (30)

Let's consider the following diagram-

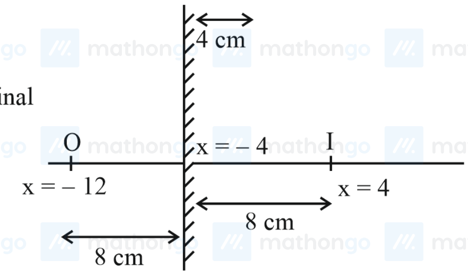


The relation between the object distance, the image distance and the radius of curvature of the spherical surface can be written as
$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R} \quad \ldots(1)$
Substitute the values of the known parameters into equation (1) and solve to calculate the required image distance.

$$\frac{1.5}{v}-\frac{1}{-15}=\frac{1.5-1.0}{30}$$
$$\Rightarrow \frac{1.5}{v}=\frac{1}{60}-\frac{1}{15}$$
$$=-\frac{1}{20}$$
$$\Rightarrow v=-30$$

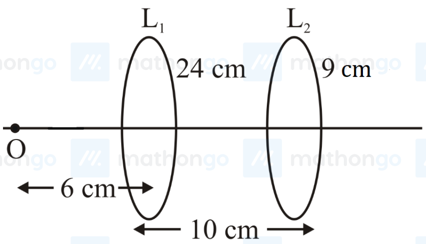
Hence, the image is formed at a distance of $30\ \text{cm}$ from the pole on the same direction as the object is placed.

8. **(2)**



When an object is placed in front of a plane mirror, object distance is equal to the image distance. In the first case, the image distance will be 12 cm from the mirror. In the next case, as the mirror is moved towards the object, the object distance becomes 8 cm. Hence, image distance will also be 8 cm.

9. **(34)**



For L_1:

$\frac{1}{v_1}+\frac{1}{6}=\frac{1}{24}$

$\Rightarrow \frac{1}{v_1}=\frac{1}{24}-\frac{1}{6}$

$=\frac{1-4}{24}$

$\Rightarrow \frac{1}{v_1}=\left(\frac{-3}{24}\right)$

$\Rightarrow v_1=-8$ cm

For L_2:

$u=-18$ cm

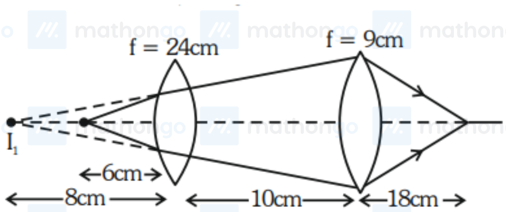
$\Rightarrow \frac{1}{v}+\frac{1}{18}=\frac{1}{9}$

$\Rightarrow \frac{1}{v}=\frac{1}{9}-\frac{1}{18}$

$=\frac{2-1}{18}$

$\Rightarrow \frac{1}{v}=\frac{1}{18}$

$\Rightarrow v=18$ cm



Distance between object and distance $=16+18=34$ cm.

10. (3)

The critical angle and the refractive index of a media are related by the formula:

$\sin \theta_c = \left(\frac{1}{\mu}\right) \quad \ldots(1)$

Substitute the value of the critical angle into equation (1) and solve to calculate the refractive index of the medium.

$\sin 45^\circ = \frac{1}{\mu}$

$\Rightarrow \frac{1}{\sqrt{2}} = \frac{1}{\mu}$

$\Rightarrow \mu = \sqrt{2}$

The refractive index of the medium can also be written as

$\mu = \frac{c}{v} \quad \ldots(2)$

Substitute the values of the known parameters into equation (2) and solve to calculate the required speed of light within the medium.

$\sqrt{2} = \frac{3\times10^8 \text{ m s}^{-1}}{v}$

$\Rightarrow v = \frac{3\times10^8 \text{ m s}^{-1}}{\sqrt{2}}$

$= 2.12 \times 10^8 \text{ m s}^{-1}$

11. (4)

The power of a lens is given by

$P_1 = \frac{1}{f}$

Using Lens Maker's formula,

$P_1 = \left(1.8 - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ and

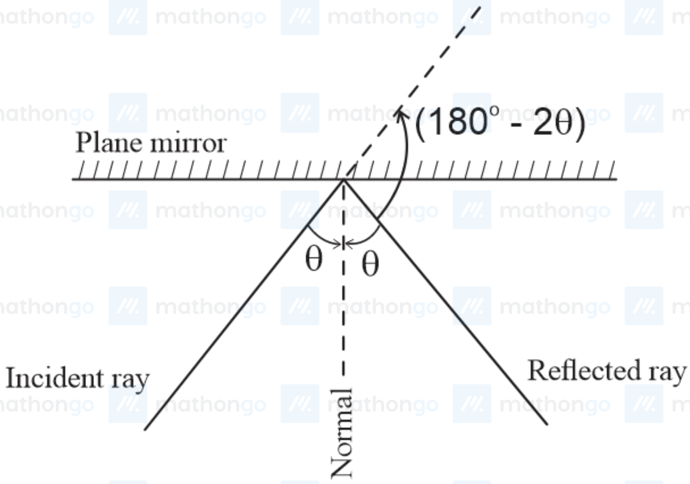
$P_2 = \frac{1}{f_{\text{immersed}}} = \left(\frac{1.8}{1.5} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Hence, the ratio is,

$\frac{P_1}{P_2} = \frac{(1.8-1)}{\left(\frac{1.8}{1.5}-1\right)} = \frac{0.8\times1.5}{0.3} = 4$

12. (1)

It is given that $i = 30^\circ$.



The formula for angle of deviation is

$\delta = 180^\circ - 2i$

Substituting the values,

$\delta = 180^\circ - 60^\circ$

$= 120^\circ$

13. (98)

The data given is

$d = 8 \text{ cm}$

$\mu = \frac{4}{3}$

Apparent depth is

$h' = \frac{d}{\mu} = \frac{8}{\frac{4}{3}} = 6 \text{ cm}$

Distance of object from mirror $= 42 + 6 = 48$ m

Distance of image from bottom of tank $= 48 + 50 = 98$ m

14. (1)

Let's assume that the original position of the bubble is x from one side of the ice cube of side a.

When the bubble is seen from one side, it can be written that

$\frac{x}{\mu} = 12 \quad \ldots(1)$

For the view from the other side, it can be written that

$\frac{a-x}{\mu} = 4 \quad \ldots(2)$

From equations (1) and (2), it is given that

$24 - 12\mu = 4\mu$

$\Rightarrow \mu = \frac{24}{16}$

$= 1.5$

15. (100)

Let L_1 be the first lens and L_2 be the second lens.

The first refraction is in L_1.

Using lens formula,

$u_1 = \infty,\ f_1 = 20\ \text{cm},$

$\frac{1}{v_1} - 0 = \frac{1}{f_1}$

$\Rightarrow v_1 = 20\ \text{cm}$

Hence, the distance between the second image and the L_2 would be $(60 - 20) = 40\ \text{cm}$.

For the second lens L_2,

$u_2 = -40\ \text{cm},\ f_2 = 20\ \text{cm},$

$\frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f_2}$

$\Rightarrow \frac{1}{v_2} + \frac{1}{40} = \frac{1}{20}$

$\Rightarrow \frac{1}{v_2} = \frac{1}{40}$

$\Rightarrow v_2 = 40\ \text{cm}$

So, the distance of image from first lens is

$x = 40\ \text{cm} + 60\ \text{cm}$

$= 100\ \text{cm}$

16. (2)

The apparent depth formula is given by $n = \frac{\textit{Real depth}}{\textit{Apparent depth}}$.

The apparent depth in this case is given by

$d' = \frac{d_1}{n_1} + \frac{d_2}{n_2}$

$\Rightarrow d' = \frac{d}{2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$

$\Rightarrow d' = \frac{d}{2}\left(\frac{n_1+n_2}{n_1 n_2}\right)$

17. (3)

Let the fish be at a depth x from the surface and let the bird be at height y from the surface. So, with respect to fish

$y' = \mu y + x$

$\Rightarrow y' = \frac{4y}{3} + x$

Differentiating with respect to time,

$\frac{dy'}{dt} = \frac{4dy}{3dt} + \frac{dx}{dt}$

$\Rightarrow \frac{4y}{3} + \frac{dx}{dt} = -12$

$\Rightarrow \frac{4}{3} \cdot \frac{dy}{dt} = -12 + 8 = -4$

$\Rightarrow \frac{dy}{dt} = -3\ \text{m s}^{-1}$

18. (5)

The focal length of the bi-convex lens (f) can be calculated as follows:

$\frac{1}{f}=\left(\mu-1\right)\left(\frac{1}{R}-\frac{1}{-R}\right)$

$=\frac{2(\mu-1)}{R} \quad \ldots(1)$

When the lens is cut into two halves, the focal length (f') of each part can be calculated as follows:

$\frac{1}{f'}=\left(\mu-1\right)\left(\frac{1}{R}-\frac{1}{\infty}\right)$

$=\frac{(\mu-1)}{R} \quad \ldots(2)$

Hence, the ratio of the powers for both the lenses can be given by

$\frac{P'}{P}=\frac{\frac{1}{f'}}{\frac{1}{f}}$

$=\frac{f}{f'}$

$=\frac{\frac{(\mu-1)}{R}}{\frac{2(\mu-1)}{R}}$

$=\frac{1}{2}$

$\Rightarrow P'=\frac{P}{2} \quad \ldots(3)$

Substitute the value of the known parameter into equation (3) to calculate the required power.

$P'=\frac{10\text{ D}}{2}$

$=5\text{ D}$

19. (30)

For minimum deviation in a prism,

$\mu=\frac{\sin\left(\frac{D_{min}+A}{2}\right)}{\sin\frac{A}{2}}$

The value of minimum deviation is

$\Rightarrow \sqrt{2}=\frac{\sin\left(\frac{D_{min}+60^\circ}{2}\right)}{\sin 30^\circ}$

$\Rightarrow \frac{1}{\sqrt{2}}=\sin\left(\frac{D_{min}+60^\circ}{2}\right)$

$\Rightarrow 45^\circ=\left(\frac{D_{min}+60^\circ}{2}\right)$

$\Rightarrow D_{min}=30^\circ$